

Quantization Erodes Rank-One Identity Geometry Before Topology

Margins, percolation, and a liveness bound for holonomy in a frozen Qwen-family capture

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[Repository](#) · [Public report and audit trail](#)

Evidence boundary. This is a preregistered, outcome-free measurement study on one frozen development capture. It tests precision-conditioned reproducibility of a rank-one representation object and the liveness of a rank-one holonomy diagnostic. It does not establish semantic identity, causal edit value, monitor sufficiency, capability preservation, or deployment safety.

Abstract

We test whether loop holonomy – gauge-invariant transport of a representation subspace around cycles in a lineage graph – detects damage caused by runtime quantization, or whether its binary topological output is inert at small scale. On a frozen Qwen-family 0.8B development capture, we measure a preregistered, target-blind, rank-one between-class object for a graph-reachability behavior family at two model sites, two weight states, and four fresh context shards. We compare NF4 4-bit double-quantized weights with float16 compute against unquantized float16 weights on byte-identical prompts. Four findings survive the registered claim boundary. First, quantization compresses the object’s support margin in all eight paired layer-19 cells, with float16-minus-4-bit point differences from 0.149 to 0.231. Second, a preregistered fine-tuning-specific sentinel is not supported: its excess precision effect is 0.041 with a descriptive 95% interval of $[-0.131, 0.244]$, indicating broad rather than fine-tuning-specific damage. Third, the registered cycle-emergence contrasts are not established; one 4-bit grid fails because a single node crosses its support gate, not because cycles disappear. Fourth, every measured cycle has positive rank-one holonomy, but this sign was mathematically forced. If edge retention is r_e and the corresponding projective principal angle is $\alpha_e = \arccos \sqrt{r_e}$, negative rank-one holonomy requires $\sum_e \alpha_e \geq \pi$. No measured cycle budget exceeds 99.7° . The binary first Stiefel-Whitney class therefore adds no evidence on this capture. A graded precursor, the frustration margin $\pi - \max_C \sum_{e \in C} \alpha_e$, is live and is eroded at layer 19 by 23.54° (22.6% of the float16 margin), meeting the registered descriptive rule. The result is a methodological correction: a stable discrete invariant is evidential only when the measured geometry permitted the alternative outcome.

1 Introduction

Mechanistic interpretability frequently treats a feature or subspace found in one context, checkpoint, or precision as the “same object” elsewhere. Similarity statistics alone do not justify that identification: squared retention discards the polar factor that transports signed coordinates, while composition around a loop can reveal path dependence that no single edge exposes. Principal-angle geometry and Grassmannian transport provide a natural language for this problem [Edelman et al., 1998, Stewart and Sun, 1990]. For a one-dimensional real subspace, loop transport reduces to a sign in $O(1) = \{+1, -1\}$. The assignment of signs to cycle classes is the first Stiefel-Whitney class w_1 of the induced real line bundle [Milnor and Stasheff, 1974].

We use runtime precision as a controlled perturbation. Low-bit quantization is operationally relevant and, in the NF4 setting, acts on weights while the prompt payload can be held byte-identical [Dettmers et al., 2023]. The experiment uses a frozen Qwen-family development model; the exact capture is identified by the sealed repository receipts rather than by a claim about a public model release. The goal is not to benchmark language-model quality. It is to ask whether a registered identity object degrades legibly under precision change and whether rank-one holonomy contributes information beyond the edge geometry from which it is computed.

The central result is partly negative and therefore useful. Quantization clearly degrades metric support, but the observed positive holonomy signs cannot be interpreted as evidence of precision stability. A simple length bound proves that every negative sign was excluded by the measured principal angles. This converts a dead categorical diagnostic into a graded liveness margin and supplies a general audit rule: before interpreting a discrete geometric invariant, verify that the measured configuration allowed the invariant to change.

2 Identity geometry and the holonomy liveness bound

2.1 Registered object and lineage graph

The measured object is a rank-one between-class direction for the registered behavior family `graph_reachability`, constructed by cross-fitted between-class scatter on frozen activations. Activations are stored in float32 in both precision arms. Candidate bases are sealed per cell before derived outcomes are read.

At a fixed site and precision, the lineage graph is a 2×4 ladder. Its nodes are weight-state by context-shard cells: two states (`base`, `naive_qlora`) crossed with four fresh context shards. The graph has eight nodes and ten edges: six adjacent-context edges within states and four checkpoint edges between states. Its cycle-space dimension is $\beta_1 = 3$, and its longest simple cycle has eight edges.

For adjacent unit directions u and v , the rank-one retention is

$$r = |u^\top v|^2 = \cos^2 \alpha, \quad \alpha \in [0, \pi/2], \quad (1)$$

where α is the principal angle between the two lines. The support statistic is the minimum edge-wise worst-direction retention. A node passes only when its support clears a matched label-permutation null boundary plus a frozen margin of 0.02. A grid is supported only when every node passes.

2.2 Rank-one transport

Choose an orientation at a starting line and transport it edge by edge using the shortest spherical lift: at each step, choose the sign of the next unit vector so that its inner product with the current vector is nonnegative. Returning to the initial line yields either the original vector or its negative. That terminal sign is the rank-one holonomy. It is invariant to arbitrary sign choices at intermediate nodes and therefore is the relevant gauge-invariant binary output.

Theorem 1 (Rank-one holonomy liveness bound). *Let $C = (\ell_0, \ell_1, \dots, \ell_m = \ell_0)$ be a cycle of one-dimensional subspaces of $\mathbb{R}^d$. Let $\alpha_i \in [0, \pi/2]$ be the principal angle between ℓ_i and ℓ_{i+1} . Under shortest-geodesic rank-one transport, negative holonomy implies*

$$\sum_{i=0}^{m-1} \alpha_i \geq \pi. \quad (2)$$

Consequently, if $\sum_i \alpha_i < \pi$, the holonomy sign is forced to be +1.

Proof. Choose a unit lift $v_0 \in \ell_0$. Recursively choose $v_{i+1} \in \ell_{i+1}$ so that $v_i^\top v_{i+1} \geq 0$. Then the spherical geodesic distance between successive lifts is $d_{S^{d-1}}(v_i, v_{i+1}) = \alpha_i$. Since $\ell_m = \ell_0$, the terminal lift is $v_m = \sigma_C v_0$ with $\sigma_C \in \{+1, -1\}$. If $\sigma_C = -1$, the path joins antipodes, whose spherical distance is π . The triangle inequality gives

$$\pi = d_{S^{d-1}}(v_0, -v_0) \leq \sum_{i=0}^{m-1} d_{S^{d-1}}(v_i, v_{i+1}) = \sum_{i=0}^{m-1} \alpha_i.$$

Thus a cycle budget below π excludes negative holonomy. □

Corollary 2 (Retention form). *If edge retentions are r_e , then*

$$\sum_{e \in C} \arccos \sqrt{r_e} < \pi \implies \sigma_C = +1. \quad (3)$$

Define the cycle frustration margin

$$F(C) = \pi - \sum_{e \in C} \arccos \sqrt{r_e}, \quad (4)$$

and the graph margin $F_{\min} = \pi - \max_C \sum_{e \in C} \arccos \sqrt{r_e}$. A positive margin means the binary sign is forced. A nonpositive margin only removes the obstruction; it does not guarantee a negative sign.

3 Registered experimental design

Three analyses were frozen and executed in order: paired precision support, edge filtration with w_1 , and a superseding frustration-margin analysis. Elementary outcomes were sealed before downstream statistics were computed.

Precision arms. The low-precision arm uses NF4 4-bit double-quantized weights with float16 compute. The comparison arm uses unquantized float16 weights. Both store captured activations in float32. This is not an fp32 arithmetic comparison.

Table 1: Registered analyses and terminal categories. The second interpretation remains in the audit trail but is superseded by the third.

Analysis	Protocol hash (abbreviated)	Terminal interpretation
Paired precision support	6437d637...adb5ef8b	Sentinel reproduced; specificity not supported
Filtration and w_1	923b2cff...9b7830aa	Positive signs observed; interpretation superseded
Frustration margin	a407503b...a53bed069	Positive signs forced by measured geometry

Prompt identity. The two precision views consume byte-identical prompts: 2,304 prompts per view, identical tokenizer source and order, and the same final non-padding-token rule. The shared payload SHA-256 begins 87d2f05e.

Cells and resampling. The design has 2 precisions $\times$ 2 states $\times$ 2 sites $\times$ 4 shards = 32 cells. Each cell uses a 256-replicate bootstrap and a 256-replicate matched permutation null. The precision-specificity contrast uses 64 inner paired replicates and 1,024 outer bootstrap replicates.

Sentinel. The preregistered sentinel is `naive_qlora` at `model.layers.19`, `shard-02`. It passes only if its complete-margin precision delta exceeds the per-draw median non-sentinel delta.

Statistical discipline. No multiplicity correction was registered. Cell intervals and max-statistic bootstrap ranges are descriptive. In particular, percentile ranges for filtration thresholds and angle budgets are not calibrated confidence intervals. The claims below distinguish exact checks, sign-consistency statements, and outcomes that merely meet a registered descriptive rule.

4 Results

4.1 Quantization compresses support margins

All eight paired layer-19 cells have positive float16-minus-4-bit complete-margin differences (Table 2; Figure 1). The point differences range from 0.149 to 0.231. Although each displayed interval excludes zero, no multiplicity-adjusted cellwise inference is claimed; the preregistered robust statement is the 8-of-8 sign consistency.

The effect is site-graded. Under 4-bit precision, layer 23 retains comfortable signed margins (0.115 to 0.220), while layer 19 lies close to the registered support boundary. One cell crosses that boundary: 4-bit, `naive_qlora`, layer 19, shard 01 has margin -0.0246 and fails its matched-label gate. The remaining 31 cells pass.

4.2 The fine-tuning-specific sentinel is not supported

The sentinel cell itself reproduces under 4-bit precision, with a positive margin of 0.061. Its excess precision effect relative to the per-draw median background, however, is 0.041 with a descriptive 95% interval of $[-0.131, 0.244]$. It therefore fails the registered specificity requirement. Within this design, precision damage is broad across states and shards rather than uniquely associated with the

Table 2: Layer-19 paired complete-margin differences. Intervals are descriptive 95% bootstrap ranges.

State	Shard	Difference	Descriptive 95% range
base	00	0.192	[0.040, 0.354]
base	01	0.197	[0.051, 0.364]
base	02	0.204	[0.047, 0.379]
base	03	0.149	[0.012, 0.306]
naive_qlora	00	0.213	[0.041, 0.404]
naive_qlora	01	0.223	[0.066, 0.377]
naive_qlora	02	0.231	[0.073, 0.422]
naive_qlora	03	0.170	[0.027, 0.336]

naive QLoRA state.

4.3 Cycle availability survives; the registered threshold shifts are not established

All four precision-by-site graphs eventually reach $\beta_1 = 3$ in the complete captured graph. Cycle emergence precedes full connectivity in every graph. The registered paired contrasts in the cycle-emergence threshold do not meet their decision rule: layer 19 has a 4-bit-minus-float16 point shift of 0.0248 with a descriptive 90% range $[-0.0037, 0.0867]$; layer 23 has a shift of 0.0108 with range $[-0.0136, 0.0551]$.

Table 3: Exact filtration thresholds. τ_{conn} is the first threshold at which the graph connects; τ_{cycle} is the first at which a cycle appears.

Precision	Site	τ_{conn}	τ_{cycle}
4-bit	layer 19	0.9059	0.9418
4-bit	layer 23	0.9159	0.9357
float16	layer 19	0.9484	0.9666
float16	layer 23	0.9182	0.9465

4.4 Positive holonomy is forced, not evidential

Each graph has seven nonzero $\mathbb{F}_2$ cycle classes: six connected simple cycles and one non-simple cycle chain. Every class has sign +1 under both construction and independent validation. Gauge preflight checks 10,752 cycle decisions across the four graphs and records zero determinant-decision mismatches.

Theorem 1 changes the interpretation. Table 4 reports the largest measured cycle budget and its complementary graph frustration margin. The maximum budget is always attained by the eight-edge outer cycle, and no point estimate exceeds 99.7° . The upper endpoints of the registered descriptive 90% bootstrap ranges also remain below 180° in all four graphs. Those ranges are not coverage-calibrated. The exact pointwise conclusion is stronger and simpler: every measured graph lies inside the forced-positive regime.

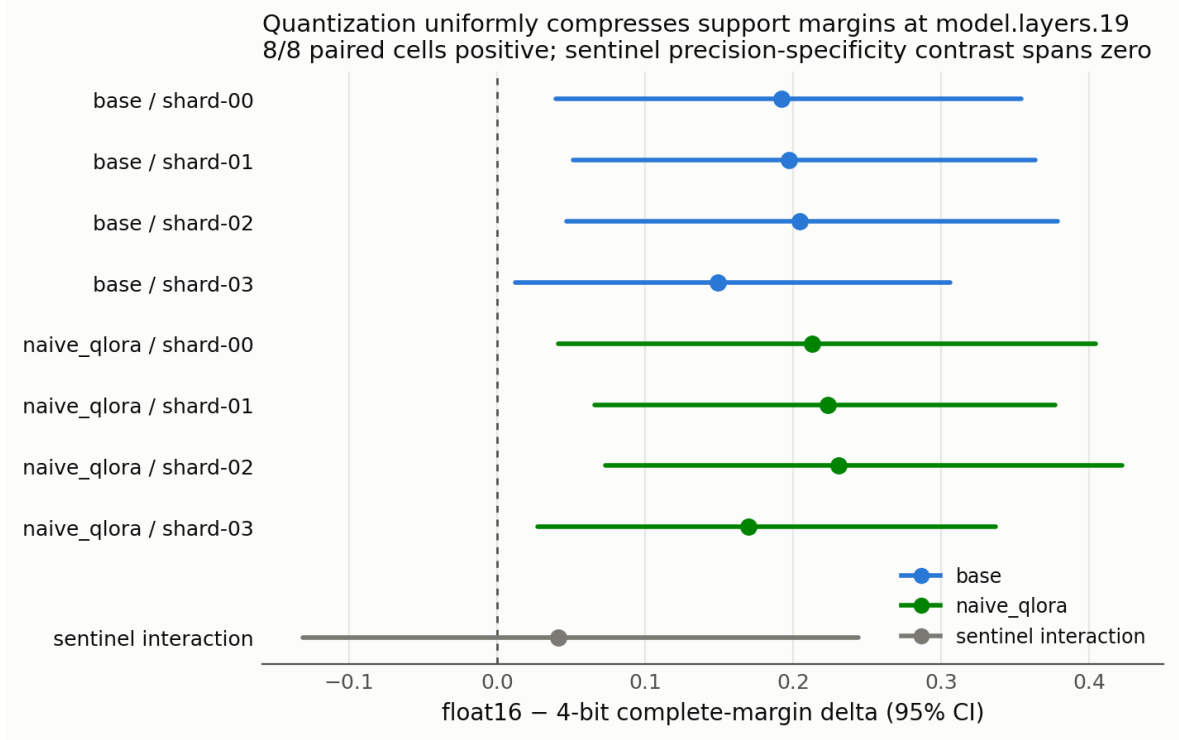


Figure 1: Float16-minus-4-bit complete-margin differences at layer 19. All eight paired point differences are positive. The intervals are descriptive and uncorrected for multiplicity.

Thus the sealed category `precision_stable Diagnostically_thin` is retained as provenance but superseded scientifically. On this capture, w_1 provides no information beyond the edge lineages that force it.

4.5 The frustration margin is a live graded precursor

Quantization increases the maximal cycle budget and therefore consumes the frustration margin (Figures 3 and 4). At layer 19, the erosion is 23.54° , equal to 22.6% of the float16 margin, with a descriptive 90% range of $[5.11^\circ, 53.11^\circ]$. This meets the registered descriptive decision rule. At layer 23, the erosion is 7.03° (8.0%) with range $[-5.20^\circ, 29.10^\circ]$ and does not meet that rule.

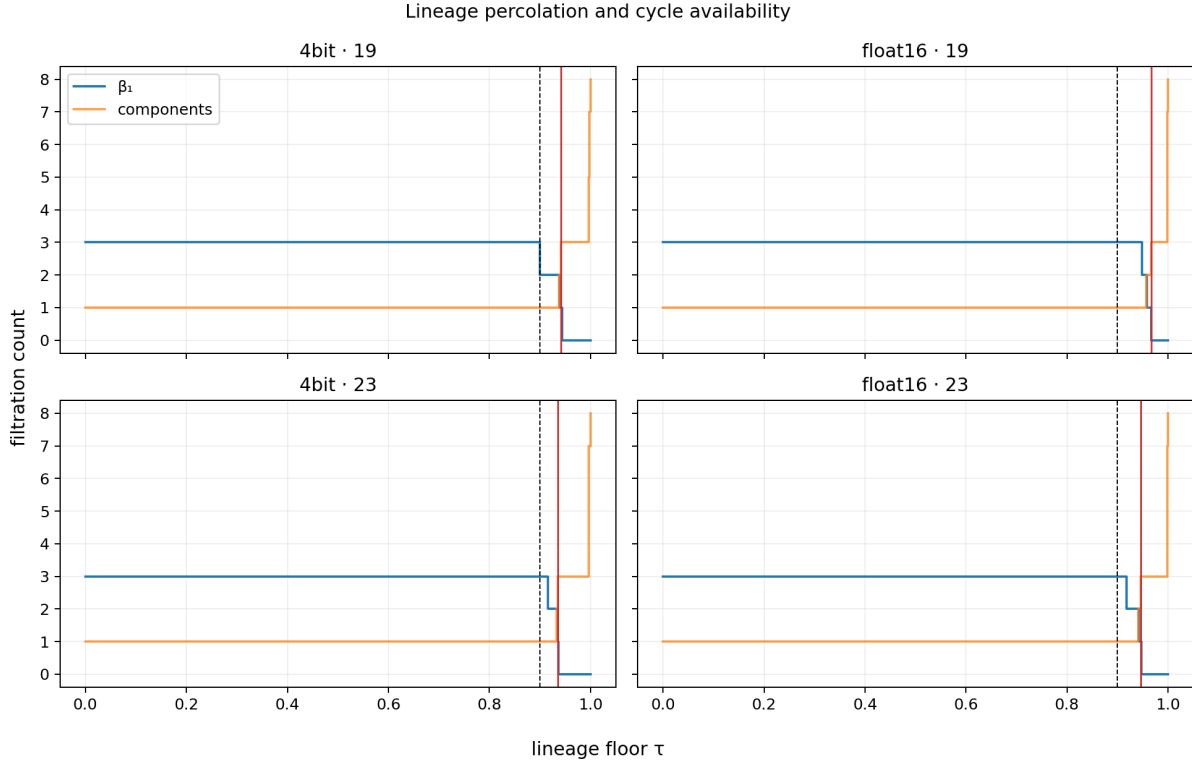


Figure 2: Edge filtration for the four precision-by-site graphs. The unsupported 4-bit layer-19 grid is attributable to a node support-gate failure, not to an absence of diagnostic cycles.

Table 4: Rank-one liveness summary. Positive graph margin means the measured geometry forces all cycle signs to be +1.

Precision	Site	Max budget	Graph margin	Upper 90% endpoint
4-bit	layer 19	99.29°	80.71°	168.49°
4-bit	layer 23	99.68°	80.32°	163.45°
float16	layer 19	75.75°	104.25°	125.82°
float16	layer 23	92.65°	87.35°	145.67°

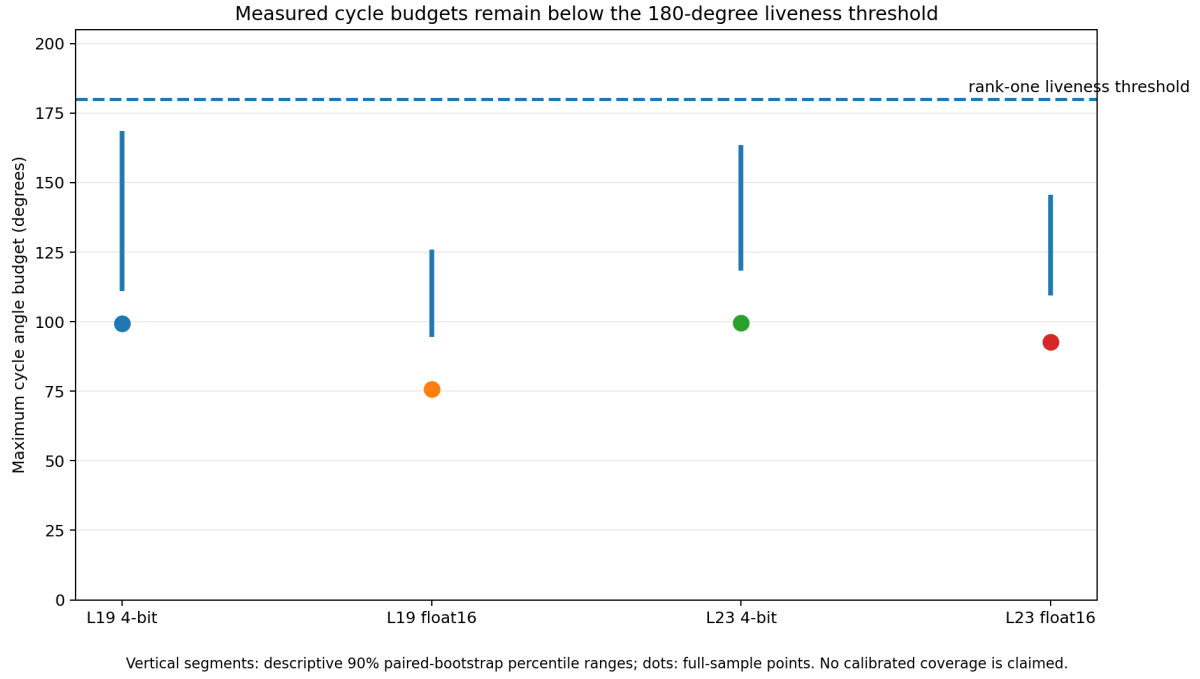


Figure 3: Maximum cycle angle budgets by site and precision. Dots are full-sample points; vertical segments are the registered descriptive 90% bootstrap ranges. The dashed line is the 180° liveness threshold. No calibrated coverage is claimed.

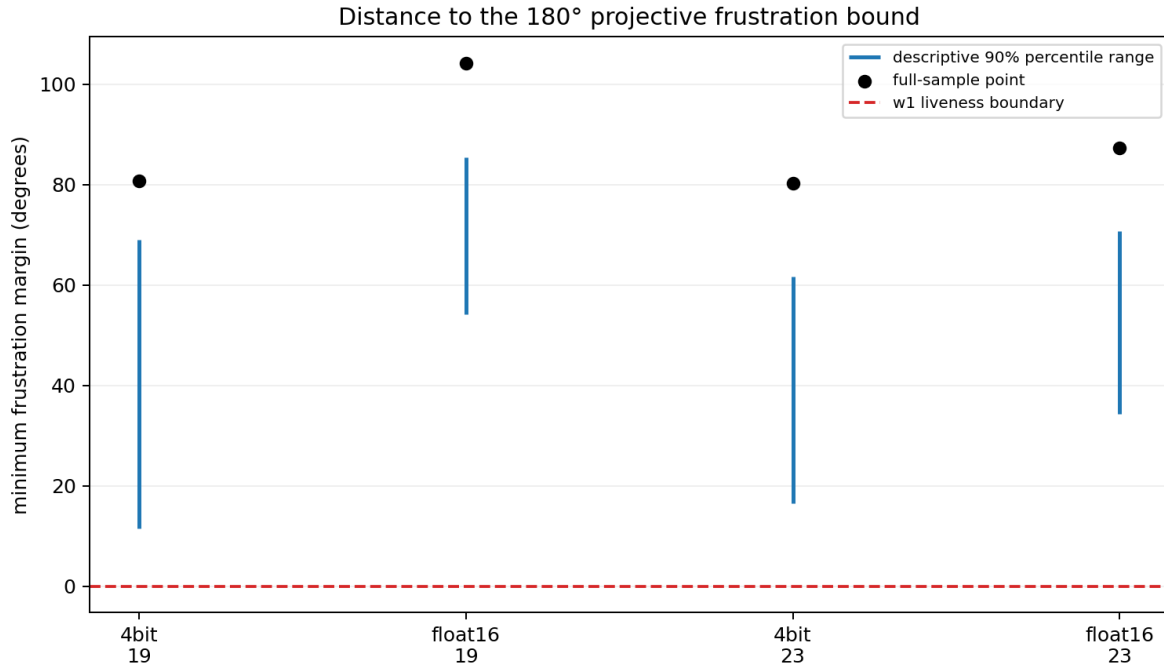


Figure 4: The same liveness geometry in margin coordinates: 180° minus the maximal cycle angle budget. Positive values lie in the forced-orientable regime. The intervals are registered descriptive 90% percentile ranges, not calibrated confidence intervals.

5 Discussion

5.1 A degradation ordering

The receipts support a strict ordering under 4-bit precision:

support margins compress $\longrightarrow$ one node gate crosses
 $\longrightarrow$ cycle angle budget grows
 $\longrightarrow$ binary topology remains unchanged.

The final arrow is not evidence of topological robustness: the alternative sign was excluded by the preceding metric geometry. In this regime, topology is the last invariant standing because the line loop is too short to become non-orientable.

5.2 The methodological lesson

A discrete geometric output should carry a liveness certificate. Reporting “trivial w_1 ” without its cycle-angle budget can mistake a deterministic consequence for an empirical confirmation. The same principle applies more broadly to thresholded or categorical diagnostics: report distance to the decision boundary and show that the experimental support included both possible outcomes.

The frustration margin is useful precisely because it preserves that distance. It is gauge-relevant, continuous in the edge retentions, and can move before the sign class becomes available. In this experiment it detects the same site ordering as the support margins while connecting the local edge damage to a global cycle-level obstruction.

5.3 A falsifiable follow-up

The obstruction to negative holonomy disappears only when a cycle budget reaches 180° ; crossing that threshold is necessary, not sufficient, for a negative sign. A follow-up should therefore be designed around liveness rather than simply adding replications to the same 2×4 ladder. Candidate axes are longer context ladders, prospectively selected weaker-lineage sites, and a controlled precision-severity sweep. The primary phase diagram should report

(cycle length, precision severity) $\mapsto$ (support margin, angle budget, holonomy sign).

The meaningful regimes are: forced positive (budget below 180°), live but positive (budget at or above 180° with sign $+1$), and frustrated (budget at or above 180° with sign -1).

6 Limitations

- One frozen development model, one behavior family, two sites, two states, and one rank-one object; no cross-model or semantic-identity claim.
- NF4 4-bit weights with float16 compute are compared with float16 weights; this is not an fp32 arithmetic study.
- The 2×4 ladder has no simple cycle longer than eight edges. Its measured geometry cannot make the binary sign test live.

- Bootstrap ranges for max-type statistics are descriptive percentiles without calibrated coverage. No multiplicity correction was registered.
- The specificity result is one preregistered sentinel contrast, not a survey of fine-tuning procedures.
- No behavioral outcomes, causal edits, or capability-preservation endpoints are consumed.

7 Reproducibility and claim boundary

The source package contains the original reports, analysis results, release manifests, and run receipts for all three analyses. The distilled release recomputes every SHA-256 named by those manifests: all 19 files match. This verifies package integrity only; it is not an independent rerun of the Qwen inference.

Table 5: Primary reproducibility identifiers.

Item	Value
Parent protocol SHA-256	6437d637d6a25efe47c462a0cadeb9aa2dd18a018a794f9e688befa4adb5ef8b
Filtration/ w_1 protocol SHA-256	923b2cff503952cc7d6eea29ff999416d0c82ee4f40f015b8804e36d9b7830aa
Frustration protocol SHA-256	a407503b908fa484e7ee09a608eed3d14f1aa7e6e3eca9e731cf34ca53bed069
Prompt payload SHA-256	87d2f05eb24e7074943b6625c907bf08362db5ffa5ac24f6a313f7d75806c14e
Prompt manifest SHA-256	9315073ff8db07fb258563957b509cdbc6ecfdc80ab0a12b869fe614077e88db
Source archive SHA-256	14317997dd4d80dff47787b0a9a48cdb21df1f8feffd90767fba34ca8b201d25

The strongest supported claim is narrow: runtime quantization degraded the registered rank-one object’s support geometry and consumed a global projective frustration margin on the frozen capture, while the binary holonomy sign remained uninformative because the measured cycle geometry forced it to be positive. The work does not authorize claims about causal control, monitor sufficiency, recursive self-improvement, or deployment safety.

Public artifacts. The report, branch history, and audit trail are available in the [RSITopology public report](#). The release bundle accompanying this PDF contains the Overleaf source, figures, selected sealed reports, and a manifest verification record.

Acknowledgment of audit history

The first filtration analysis interpreted the all-positive signs as precision-stable but diagnostically thin. The superseding analysis showed that positivity was forced. Both registrations are retained. This correction is part of the result: it demonstrates why liveness checks belong upstream of categorical interpretation.

References

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